Expanded engineering and reliability release gates

Status: Review-ready release-control specification. It expands, but does not prove satisfaction of, the existing release-gate document.

1. Release object and evidence pack

Every promotion has one immutable `release_id` with artifact hash, environment, tenant/domain scope, feature-flag state, schema/mapping/configuration versions, approvers, evidence links, rollback plan, and expiry. A deployment is not an activation; an activation is not institutional authority.

2. Gate matrix

Gate	Required evidence	Blocking conditions	Accountable approver
Scope & authority	Domain authority, tenant sponsor, intended users, permitted commands, feature flags	Ambiguous record authority; unsupported customer claim	Product + institutional owner
Tenant isolation & authorization	Automated deny tests for routes, RPCs, DB/RLS, storage, search, exports, jobs, caches	Any cross-tenant read/write; unscoped privileged path	Engineering + security
Data/provenance	Source, authority, freshness, classification, retention, migration/reconciliation evidence	Unknown source/mapping; stale or unreconciled official data	Domain owner + data owner
Security & privacy	Threat/risk review, secret/dependency scans, vendor/AI route approval, privacy impact as needed	Critical unresolved risk; prohibited data flow; expired exception	Security/privacy owner
Accessibility	Complete-process keyboard and assistive-technology evidence for default/loading/empty/error/denied/offline states	Blocker in a critical journey; no non-AI/non-drag alternative where required	Accessibility reviewer
Functional & reliability	Unit, contract, integration, E2E, concurrency/idempotency, failure-mode and recovery evidence	Missing high-risk test; non-deterministic official transaction	Engineering owner
Observability & support	SLI/SLO definition, dashboards, alerts, correlation, sanitized telemetry, runbook, on-call/support owner	Alert without action owner; sensitive telemetry leakage; no escalation path	SRE/support owner
Rollback & recovery	Tested rollback/compensation, backup/restore relevance, cutover checkpoint, communications plan	Unrecoverable change without authorized exception	Engineering + domain owner
Institutional activation	Tenant configuration/policy approval, training, support and integration readiness, reconciliation	No tenant approval; unverified external write/receipt	Institution owner

Gate	Required evidence	Blocking conditions	Accountable approver
Claims & documentation	Release notes, customer-facing wording, known limits, support content, evidence state	Statement implies verification/certification not shown	Product + legal/comms reviewer

3. Stage transition rules

Designed → Implemented

Architecture, authority, policy, data contracts, and threat/access review are recorded. No claim of operational behavior.

Implemented → Tested

Automated and manual results identify artifact/environment and prove the scoped behavior, including expected denials and failure paths.

Tested → Deployed

A controlled artifact is deployed disabled or in a protected test environment, with configuration and migration manifest captured.

Deployed → Institution-authorized

The accountable institution accepts the exact tenant, domain, data, roles, integrations, and operational model. This authorization does not enable a different tenant or broader authority.

Institution-authorized → Production-enabled

The feature is enabled only after all relevant gates pass for the exact activation scope, real monitoring/support are ready, and any authoritative flows reconcile successfully.

4. High-impact commands

Writes that register, grade, share, export, pay, alter access, delete, or affect official records must use request → exact preview → authorized decision → execute → receipt/reconciliation. Approval binds to target, content, scope and expiry; any material change invalidates approval.

5. Go/no-go and exception control

- Each gate has pass/fail/not-applicable status, evidence, reviewer and expiry.
- “Not applicable” must state why and be approved by the relevant owner.
- Exceptions name risk, compensating control, scope, owner, expiry, monitoring and reversal condition. Exceptions never waive tenant isolation, prohibited data egress, or critical official-record integrity defects.

- A failed critical gate is a no-go. Lower-risk acceptance must be explicit and cannot create misleading public claims.

6. Post-activation checks

At defined checkpoints, verify authorization denials, error/latency/freshness signals, source receipts, user impact, support volume, accessibility reports, reconciliation, budget use, and rollback readiness. Disable by tenant/domain/feature flag on material risk; preserve evidence and conduct remediation before re-enable.